TOP 50 DEFINITIONS

Ethereum & Sepolia — Practical Exam / Viva Guide

Format: Every topic is indexed (1–50) and includes a short definition plus a practical example.

Revision priority: Focus especially on Ethereum, Sepolia, Sepolia ETH, smart contracts, Solidity, Remix IDE, deployment, ABI, gas, transaction hash, MetaMask, and Etherscan.

1. Blockchain

Definition: A distributed digital ledger that records transactions in blocks linked using cryptographic hashes.

Example: A cryptocurrency transfer can be recorded in a block and shared across the network.

2. Block

Definition: A data structure containing a group of transactions and information linking it to the previous block.

Example: A block may contain several ETH transfers made during a particular period.

3. Node

Definition: A computer running blockchain software that communicates with the network and may verify or store blockchain data.

Example: A computer running an Ethereum client and checking transactions is an Ethereum node.

4. Decentralization

Definition: The distribution of control across multiple network participants instead of one central authority.

Example: Ethereum is maintained by many independent nodes and validators.

5. Distributed Ledger

Definition: A shared record of transactions maintained and synchronized across multiple computers.

Example: Different Ethereum nodes maintain copies of blockchain data.

6. Hash

Definition: A fixed-length cryptographic output generated from input data, used to identify data and detect changes.

Example: A block header is hashed to produce its block hash.

7. Consensus Mechanism

Definition: A protocol that allows blockchain participants to agree on the valid state and order of transactions.

Example: Ethereum uses Proof of Stake to reach consensus.

8. Ethereum

Definition: A decentralized, programmable blockchain platform that supports smart contracts and decentralized applications.

Example: A developer can build a token or voting application on Ethereum.

9. Ether (ETH)

Definition: The native cryptocurrency of Ethereum, used to pay transaction fees and support network activities.

Example: A user pays ETH to submit a transaction on Ethereum Mainnet.

10. Ethereum Mainnet

Definition: The primary Ethereum network where real transactions take place using real ETH.

Example: Deploying a contract to Mainnet can involve real financial costs.

11. Testnet

Definition: A separate blockchain network used to develop and test applications without risking real mainnet funds.

Example: A developer tests a new contract on Sepolia before using Mainnet.

12. Sepolia

Definition: An Ethereum test network used primarily to test smart contracts and decentralized applications.

Example: A student deploys a sample Solidity contract on Sepolia for a practical exam.

13. Sepolia ETH

Definition: Test cryptocurrency used to pay gas fees on Sepolia; it has no intended monetary value.

Example: A developer requests Sepolia ETH from a faucet to deploy a test contract.

14. EVM (Ethereum Virtual Machine)

Definition: The execution environment that runs Ethereum smart-contract bytecode and applies state changes.

Example: The EVM executes a contract function that updates a stored message.

15. Smart Contract

Definition: A program deployed on a blockchain that executes according to predefined rules when invoked.

Example: A voting smart contract records a vote and prevents a user from voting twice.

16. Solidity

Definition: A high-level, statically typed programming language commonly used to write Ethereum smart contracts.

Example: A developer writes a HelloWorld contract using Solidity.

17. Remix IDE

Definition: A browser-based development environment used to write, compile, deploy, and test Solidity smart contracts.

Example: A student compiles and deploys HelloWorld using Remix IDE.

18. Contract Deployment

Definition: The process of publishing compiled smart-contract code to a blockchain through a contract-creation transaction.

Example: Clicking Deploy in Remix with Sepolia selected sends a deployment transaction.

19. Contract Address

Definition: The unique blockchain address assigned to a deployed smart contract.

Example: An application uses the contract address to call the deployed HelloWorld contract.

20. ABI (Application Binary Interface)

Definition: A machine-readable description of a contract's functions, inputs, outputs, and events, used by applications to interact with it.

Example: A frontend uses the ABI to call the contract's getMessage() function.

21. Bytecode

Definition: The low-level instructions generated by compiling a smart contract, which the EVM executes.

Example: The Solidity compiler converts HelloWorld.sol into EVM bytecode.

22. Function

Definition: A named block of code in a contract that performs an operation and may read or modify data.

Example: A setMessage() function updates a message stored in a contract.

23. Constructor

Definition: A special Solidity function that runs once when a contract is deployed to initialize its state.

Example: A constructor sets the initial message to 'Hello'.

24. State Variable

Definition: A variable declared in a contract whose value is stored persistently on the blockchain.

Example: A string public message stores the contract's current message.

25. Event

Definition: A structured log emitted by a smart contract so external applications can detect that an action occurred.

Example: A Transfer event is emitted when tokens are transferred.

26. Transaction

Definition: A signed instruction submitted to Ethereum, such as transferring ETH, deploying a contract, or invoking a state-changing function.

Example: Sending 0.01 ETH to another wallet creates a transaction.

27. Transaction Hash (TxHash)

Definition: A unique identifier for a transaction that can be used to look up its details in a blockchain explorer.

Example: Paste the transaction hash into Sepolia Etherscan to inspect the transaction.

28. Gas

Definition: A unit that measures the computational work required to execute operations on Ethereum.

Example: Executing a contract function consumes gas.

29. Gas Fee

Definition: The amount paid for the gas consumed by a transaction, calculated using gas used and the applicable gas price.

Example: A Sepolia transaction uses test ETH to pay its gas fee.

30. Gas Limit

Definition: The maximum amount of gas a sender allows a transaction to consume.

Example: A wallet estimates a gas limit before sending a contract transaction.

31. Gwei

Definition: A denomination of ETH commonly used to express gas prices; one Gwei equals 0.000000001 ETH.

Example: A displayed gas price of 20 Gwei means 20 billionths of an ETH per gas unit.

32. Wallet

Definition: Software or hardware that manages cryptographic keys and helps users view accounts, sign transactions, and connect to blockchain applications.

Example: MetaMask lets a user connect to Sepolia and approve a transaction.

33. MetaMask

Definition: A self-custody cryptocurrency wallet used to manage Ethereum-compatible accounts and connect to decentralized applications.

Example: A user selects Sepolia in MetaMask before deploying a contract from Remix.

34. Public Key

Definition: A cryptographic key that can be shared and is mathematically related to a private key.

Example: A digital signature can be verified using the corresponding public key.

35. Private Key

Definition: A secret cryptographic value used to create digital signatures and authorize actions from an account.

Example: A wallet uses its private key to sign a transaction; the key must never be shared.

36. Digital Signature

Definition: A cryptographic proof created using a private key that allows others to verify authorization.

Example: MetaMask signs a transaction before it is broadcast to Ethereum.

37. Faucet

Definition: A service that distributes small amounts of test cryptocurrency for development and testing.

Example: A developer uses a Sepolia faucet to receive test ETH.

38. Blockchain Explorer

Definition: A tool or website that lets users inspect blockchain blocks, transactions, addresses, and smart contracts.

Example: Sepolia Etherscan can show whether a test transaction succeeded.

39. Etherscan

Definition: A blockchain explorer for Ethereum that displays transactions, addresses, blocks, tokens, and contract information.

Example: A user searches a transaction hash on Sepolia Etherscan.

40. dApp (Decentralized Application)

Definition: An application that uses smart contracts or decentralized infrastructure for some of its core functionality.

Example: A web-based voting interface that sends votes to a smart contract is a dApp.

41. Web3

Definition: A broad term for internet applications built around decentralized networks, blockchain, smart contracts, and digital assets.

Example: A user connects a wallet to a Web3 application instead of creating a conventional username and password account.

42. NFT (Non-Fungible Token)

Definition: A unique blockchain token that is distinguishable from other tokens and can represent a particular digital item or claim.

Example: An artist mints a unique digital artwork as an NFT.

43. ERC-20

Definition: An Ethereum token standard defining a common interface for fungible tokens, including balances and transfers.

Example: A token contract implementing ERC-20 can represent a fungible campus reward token.

44. ERC-721

Definition: An Ethereum token standard commonly used for non-fungible tokens, where each token has a distinct identifier.

Example: A digital certificate can be represented by an ERC-721 token.

45. Proof of Stake (PoS)

Definition: A consensus mechanism in which validators stake cryptocurrency and participate in proposing or attesting to blocks.

Example: Ethereum validators stake ETH to help secure the network.

46. Validator

Definition: A participant in Ethereum Proof of Stake that runs validator software and performs duties such as proposing or attesting to blocks.

Example: An Ethereum validator attests to a proposed block.

47. RPC Endpoint

Definition: A network address through which an application sends requests to an Ethereum node to read data or submit transactions.

Example: Remix or a wallet can use an RPC endpoint to connect to Sepolia.

48. Chain ID

Definition: A numeric identifier for a blockchain network, used by wallets and applications to distinguish networks.

Example: Sepolia has its own chain ID, separate from Ethereum Mainnet.

49. Transaction Receipt

Definition: A record returned after a transaction is included and executed, containing details such as execution status, gas used, and logs.

> **Example:** A receipt showing success confirms that a deployment transaction executed successfully.

50. Read-only Call (eth_call)

Definition: A way to execute a contract function to read blockchain state without submitting a state-changing transaction or paying an on-chain transaction fee.

> **Example:** Calling getMessage() in Remix reads the stored message without changing the contract.

Quick practical sequence

Write a Solidity contract → compile it in Remix → connect MetaMask → select Sepolia → obtain Sepolia ETH from a faucet → deploy the contract → copy its contract address → inspect the deployment transaction on Sepolia Etherscan.

Security reminder: Never share your private key or Secret Recovery Phrase. Testnet ETH is for testing and has no intended monetary value.